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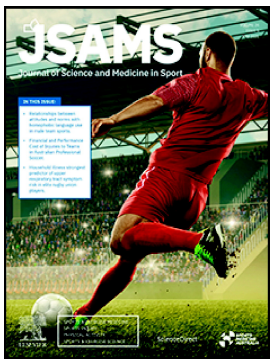


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# Neurocognitive function and musculoskeletal injury risk in sports: a systematic review

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**Abstract**

**Objectives:** A variety of modifiable risk factors have been suggested to predict musculoskeletal injury in sports. Screening paradigms and exercise-based prevention programs, so far, frequently focused on physical parameters such as strength, balance or flexibility. However, less is known with regard to the impact of brain function. We investigated the relation between cognition and the occurrence of injury in athletes. **Design:** Systematic review. **Methods:** Database searches using Pubmed, Web of Science and Google Scholar returned nine cohort and case-control studies, which almost exclusively examined college athletes engaging in open skill sports. Methodological quality, rated by means of a modified Downs and Black checklist, was moderate (12/17 points). **Results:** Seven out of the nine included studies suggested that lower-order cognitive function (i.e. visuomotor reaction time) is systematically associated with the occurrence of injury. No paper examined the impact of executive function. **Conclusions:** Despite initial evidence pointing towards a potentially injury-predictive role of lower-order cognitive function, several issues including the use of more variable populations and outcomes remain to be addressed before definitive recommendations for clinicians can be made.

**Key words:** cognition, sports injury, trauma, brain, athletes

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## Introduction

Musculoskeletal injuries are a major health problem in sports and considerable efforts have been made to gain insight into the aetiology of related complaints. According to the available evidence, causative factors can be numerous, inter alia including age, sex, skill level, physical fitness, training load, or joint biomechanics.[1] The increasing knowledge about potential predictors has also resulted in the development of prevention strategies aiming to reduce injury risk. Hitherto, movement-based approaches (e.g. strength, balance, power, or flexibility training[2-4]) mainly focus on bodily aspects of movement requiring little or no cognitive control. This, however, neglects that sporting activity can substantially depend on the rapid adaptation to variable situational needs. In open skill sports such as football, handball, basketball, or soccer, athletes engage in an unpredictable environment requiring spontaneous reactive decision-making,[5] e.g. when dodging tackles or sidestepping defenders.

Although there is no universally accepted classification, many researchers have proposed a general distinction between higher-order and lower-order cognitive functions.[6-8] Lower-order skills such as visual scanning, reaction time, processing speed, attention, and short-term memory represent mostly automated processes with limited complexity. In contrast, higher-order or executive functions draw on these rather basal skills to allow dynamic and goal-directed behavior depending on the situational needs.[6-8] Executive function is typically suggested to consist of three components. First, working memory describes the ability to mentally combine or split stored information. It hence goes beyond the simple recall of memories.[7,8] As an example, a striker in football needs to integrate knowledge about the positions of teammates and opponents as well as their current and previous running speeds/trajectories to find the optimal passing path. Second, inhibitory control has two relevant sub-domains. On the attentional level, it helps to focus on certain thoughts, actions or processes without being distracted while the behavioral level includes the ability to stop automated/ prepotent actions.[9] The striker originally planning to pass the ball to a teammate may need to spontaneously cancel his plan in the last moment when noticing an approaching defender. Third, cognitive flexibility allows efficient task switching and flexible adaptation to new situations.[7,8] It would be needed if the striker decides to side-step the defender with a quick change of direction after having interrupted the aforementioned passing maneuver. Similar to the brain, which has been shown to work in complex

networks,[9] the various cognitive functions never act in isolation. For instance, switching a motor plan (cognitive flexibility) requires both, effective canceling of the current action (inhibitory control) as well as the recall of possible alternative movements which depend on knowledge about the environment (working memory).

Besides its contribution to sports performance, cognitive function (i.e., EF) has also been suggested to help prevent injuries.[10] This assumption is based on the idea that musculoskeletal traumas are rarely observed under controlled and pre-planned closed skill conditions, e.g. while walking, running or cycling. Contrarily, it may be argued that injury risk is substantially higher if open-skill activities, requiring spontaneous adaptations to a dynamic environment, are performed. Systematic analyses describing the most frequent mechanisms of injury in sports support this theory, showing that most traumas occur during duels (e.g. tackling/sidestepping) and dual-task actions (e.g. throwing while jumping).[11-13] In these situations, the brain is required to integrate an enormous amount of sensory input within shortest time in order to ensure successful decision-making and/or adequate reactive motor adaptations.[10] Interestingly, a plethora of studies suggest that athletes with a history of concussion and concomitant decreases in neurocognitive function have twofold higher odds to incur a musculoskeletal injury than non-concussed individuals.[14] However, it is unclear as to whether these findings can be extrapolated to cognitive high- and low performers without previous concussion. The present systematic review therefore investigated the association between cognitive function and injury occurrence in non-concussed athletes.

## Methods

A systematic review was performed, adhering to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines[15] and following the recommendations for ethical publishing of systematic reviews proposed by Wager and Wiffen.[16] Prior to initiation, the study was registered in the PROSPERO database (CRDXXXXXXX).

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In March 2021, two independent investigators (JW, DG) performed a systematic literature search. Articles pertaining to the research question were identified using MEDLINE (PubMed), Web of Science and Google Scholar. The terms for all databases were similar but modified according to the requirements of the respective search masks. As an example, the term for PubMed was: 'neurocog* AND "injur* AND (risk OR predict) AND athletes'. For Google Scholar, an approach used in earlier systematic reviews of our group[5,17] was applied. Briefly, the first 100 hits displaying the most relevant findings with reference to the search term were checked for eligible articles in order to complement the results of the other database searches. Furthermore, the reference lists of all included papers were checked to identify further potentially relevant studies.[18]

Prospective or retrospective observational studies (case-control or cohort design) were considered for inclusion. Additional criteria were (1) enrolment of healthy individuals, (2) regular engagement of study participants in sports (no restriction of type or expertise level), (3) assessment of neurocognitive function, (4) systematic documentation of injury incidence, (5) publication in English language in a peer-reviewed journal. Studies including participants with a history of concussion or severe musculoskeletal injury as well as studies testing the effects of interventions were excluded. Following the recommendation of Scherer & Saldanha,[19] both, full articles and conference abstracts were included.

Using a standardised assessment sheet, two investigators (JW, DG) independently extracted the following data from included papers: study design, sample size, participant characteristics, cognitive outcomes, injury incidence, measured associations/differences.

We evaluated the methodological quality of the included studies (available full-texts) using a modified Downs and Black checklist, which has been shown to be a reliable tool for rating of observational studies [20]. Ratings including categories on reporting (6 items), external validity (2 items), internal validity – bias (2 items), internal validity – confounding (2 items), and power (1 item) were made, yielding a sum score with a maximum of 17 points. Ratings were performed by two independent investigators (JW, DG), resolving disagreements by discussion.

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## Results

The flow of the literature search is displayed in **Fig. 1**. Collectively, the three databases returned a total of 404 studies. After removal of duplicates and application of the inclusion criteria, nine studies, seven with a cohort design[21-27] and two with a case-control design,[28,29] were considered eligible (**Tab.1**). Except for one article examining adolescents in skiing and snowboarding,[24] all other studies included adult athletes engaging in open skill sports such as soccer or football. Outcomes exclusively consisted in tools assessing lower-order cognitive functions. Averaged over the total sample, methodological study quality was moderate (12/17 points, range: 10-15 points, **Tab. 2).**

With regard to cohort studies, Taimela et al.[21] measured baseline visuomotor reaction time in a sample of 37 soccer players. Stepwise logistic regression revealed an association of a longer time needed to complete the assessment and the occurrence of injuries during the prospective follow-up period. Similar findings were made by Dvorak and colleagues.[22] The authors performed assessments of reaction time in 264 soccer players, which were prospectively monitored for one year. Athletes sustaining an injury (+18%) or severe injury (at least 4 weeks downtime, +24%) demonstrated significantly higher baseline reaction times than athletes remaining injury-free. Wilkerson[23] prospectively monitored 76 college football players for one season, observing 29 lower extremity sprains and strains. The cut-off of ≥550 ms in the reaction time baseline assessment had 74% sensitivity and 51% specificity to distinguish injured from non-injured athletes (Odds ratio: 2.94). In another, very similar prospective study, Wilkerson and colleagues[25] obtained a slightly higher cut-off (≥ 705 ms) with 67% sensitivity and 54% specificity. Athletes with slower reaction time presented 2.3 fold higher odds of being injured during the follow-up period. McDonald et al.[27] used the ImPACT battery to examine a variety of skills including reaction time, processing speed and memory in football players monitored prospectively over one or two seasons. While reaction time was a predictor of lower limb injury (adjusted odds ratios of 4.78 for season 1 and 2.57 for season 2), no clear relationship was found for the other outcomes despite some trends and uncorrected univariate associations. Similar to McDonald et al.,[27] Faltus et al.[24] used baseline ImPACT battery scores to compare alpine ski and snowboard athletes with and without injury sustained during a three-year period (retrospective approach). No differences in lower-order cognitive functions were identified between cohorts. Ha et

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al.[26] included college athletes from a variety of sports (basketball, rugby, baseball, ice hockey, soccer). Cognitive function was tested using the Standardized Assessment of Concussion (SAC). During the six-month prospective follow-up period, 14 musculoskeletal injuries were registered in the 77 included participants. Data analysis did not reveal an association between cognitive function and the occurrence of trauma.

With regard to case-control studies, Swanik et al.[28] used pre-season baseline performance in a testing battery capturing lower-order cognitive functions (ImPACT) to compare college athletes (various sports) with anterior cruciate ligament injury and matched controls. Individuals with sustained traumas scored lower in the composite score as well as in the sub-domains verbal memory, visual memory, processing speed, and reaction time. Eyerly et al.[29] matched high school athletes with and without ACL rupture (5:1 ratio). While they did not identify between-group differences for visual memory, injured individuals had worse baseline performance in reaction time, processing speed and verbal memory.

## Discussion

Despite its arguable relevance in sports, data describing the relationship between neurocognitive performance and the occurrence of musculoskeletal injury are still scarce. The majority of the available evidence, however, suggests that deficits in brain function (i.e. visuomotor reaction time) could predispose athletes to sustaining a trauma. This finding fits with a previous systematic review showing an association between the level of cognitive skills and the functional and/or biomechanical safety of movement incorporating time-constrained decision-making.[30] Specifically, it has been shown that individuals with lower cognitive skills exhibit higher ground reaction forces,[31] knee abduction moments,[31] tibial anterior shear forces,[31] center of pressure sway[32] as well as smaller Hamstring-Quadriceps ratios[33] during unanticipated jump-rebound or jump-cut tasks. Although Almonroeder[34] did not make such observation, it may, in sum, be assumed that the effect of cognitive function on injury risk could be mediated by alterations in movement biomechanics.

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The potential predictive value of cognitive function with regard to musculoskeletal injury risk may particularly be of value for physiotherapists and conditioning coaches considering the development and implementation of specific cognitive testing and training paradigms. So far, the movement-related diagnostic spectrum mainly comprises assessments of physical function with limited cognitive contribution.[2-4] Possibly, a more multifactorial approach in risk screening could increase the chance to successfully predict injury. However, before definitive recommendations can be made, several gaps in the literature need to be addressed.

All examined studies measured lower-order cognitive functions. Related measures such as attention reaction time, processing speed, or memory may play a significant role for athletes' health. Quick processing of sensory input represents the basis for motor actions initiated to prevent a trauma and the time between perception of a stimulus and the initiation of an adequate reaction is crucial.[10] In an exploratory study, Diekfuss et al.[35] used functional magnet resonance imaging to examine female high school soccer players who later sustained an ACL rupture. Compared to athletes without subsequent trauma, they exhibited lower levels of functional connectivity between the left primary sensory cortex and the right posterior lobe of the cerebellum, which is paramount for motor coordination. This may suggest that individuals at risk of injuries, in some instances, cannot forward relevant proprioceptive information needed to correct a current motor action in time.

Notwithstanding, despite the importance of lower-level cognitive functions, it could be assumed that executive functions have a similar or higher relevance. Arguably, the appropriateness of a motor reaction to an external stimulus will depend on more complex higher-order skills such as working memory or task-switching. Huijgen et al.[36] investigated the possible link between the expertise level in soccer and a variety of cognitive skills. While no association was found for lower-order functions, executive functions, namely cognitive flexibility and inhibitory control, clearly discriminated elite and sub-elite players. As decision-making particularly requires executive skills, a similar constellation seems plausible regarding injury risk. Associations between cognition and the odds to sustain a trauma may hence be larger for inhibitory control, working memory or cognitive flexibility than for simple skills such as reaction time or processing speed and future studies with related outcomes are urgently warranted.

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Seven out of nine studies revealed possible associations of cognitive function and injury occurrence, but two studies did not. This appears to indicate that cognition may not equally contribute to risk prediction in all populations. Contrarily to the other researchers examining adults in interactive team sports (e.g. soccer, football, basketball), Faltus et al.[24] focused on adolescents engaging in skiing and snowboarding. While the latter two activities definitely require fast adaptations to the environment, they do not entail the need to directly react to other competing individuals or opponents and with this, different cognitive contributions. This distinction, however, does not apply to the trial of Ha et al.[26] who had similar participants and outcomes to those studies detecting a role of cognition. This may be due to two factors. Firstly, it is notable that only 14 injuries occurred during the short follow-up period of six months. Questions, moreover, arise regarding the definition of injury. One of the registered complaints was a cramp, which is unlikely to be prevented by cognitive skills. In sum, it is unclear, under which circumstances brain function relates to the occurrence of injury and the hypothesis that it is of higher relevance in open-skill sports requires further verification.

Finally, although first indications for a link between cognitive function and injury risk exist, there is no evidence for the success of preventive exercise interventions addressing the deficits. A few studies showed that performance in outcomes of cognitive function can be improved. For instance, Yotani et al.[37] examined the effects of an 8-week stimulus-response training in athletes from a variety of sports, detecting a 17%-decrease of visuo-motor response time in the intervention group. Following their prospective study phase, Wilkerson et al.[25] submitted their participants with poor cognitive baseline performance to a 6-week device-based visuomotor response training. Contrarily to an inactive control group, a significant reduction (~25%) was found in post-measurements of reaction time. Whilst training intervention seem capable of modifying markers of cognitive function, it is unclear if these near-transfers will translate into a reduction of injuries. Further research is therefore needed in order to conclusively judge the potential value of preventive training interventions with regard to the risk to sustain a musculoskeletal trauma.

## Conclusion

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Initial evidence suggests that cognitive function (i.e. visuomotor reaction time) may be associated with the occurrence of musculoskeletal trauma in sports. However, additional research with prospective designs, varying populations, rigorous injury definitions, and assessments of higher-order executive functions is urgently needed before definitive conclusions regarding its value in the prevention injuries can be drawn.

**Practical implications**

- Reaction time seems to be a relevant predictor of musculoskeletal injury in open-skill sports
- Exercise professionals may consider developing screening measures and preventive screening measures addressing neurocognitive function
- More research regarding the impact of executive function is needed

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| Study | Design | Population | Sample | Outcome | Result |
|---|---|---|---|---|---|

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**Tab.1.** Characteristics of the included studies.

| | | | | | |
|---|---|---|---|---|---|
| Taimela et al.[21] | Cohort (P) | Male soccer players | n=37 athletes | RT (button press in response to light stimulus), injury incidence (one year) | Higher RT predictive of musculoskeletal injury*. |
| Dvorak et al.[22] | Cohort (P) | Soccer Players (sex unknown) | n=264 athletes | RT (instrument not described), incidence of injury (one year) | Injured (128 ms) and severely injured players (134 ms) with higher RT than uninjured players (108 ms), no OR reported |
| Swanik et al.[28] | Case-Control (R) | Male and female college athletes (diverse sports) | 80 cases with ACL rupture, 80 matched controls | ImPACT (visual memory, verbal memory, visual motor speed, RT), incidence of ACL rupture (three seasons) | Higher RT (+7%), lower processing speed (-9.8%) as well as lower verbal (-4.8%) and visual (-13.9%) memory scores in injured athletes |
| Wilkerson et al.[23] | Cohort (P) | College football players (sex unknown) | n=76 athletes | Visuomotor RT component of ImPACT scale, incidence of lower extremity sprain/strain (one season) | Low-performers (≥545 ms) with OR of 2.94 and RR of 2.17 for being injured. |
| Faltus et al.[24] | Cohort (R) | Male and females adolescent skiing and snowboard athletes | n=134 athletes | ImPACT (visual memory, verbal memory, visual motor speed, RT), incidence of injury (three seasons) | No differences between injured and non-injured athletes. |
| Wilkerson et al.[25] | Cohort (P) | College football players | n=76 athletes | Visuomotor RT (Dynavision D2 system), incidence sprains/ strains injury (one season) | Low-performers (≥705 ms) with OR of 2.30 for being injured. |
| McDonald et al.[27] | Cohort (P) | College Football players | n= 142 athletes | ImPACT (visual memory, verbal memory, visual motor speed, RT), incidence of injury (one or two seasons) | Low performers (RT ≥685 ms for season 1 / ≥800 ms for season 2) with adjusted OR of 4.78 (season 1) / 2.57 (season 2) for being injured. Despite univariate associations, no significant adjusted OR for processing speed and memory. |
| Ha et al.[26] | Cohort (P) | Male elite college athletes (diverse sports) | n=77 athletes | Standardized Assessment of Concussion (SAS; mental test, immediate memory, delayed memory, concentration), incidence of lower extremity injury (6 months) | No association of cognitive performance and injury occurrence. |
| Eyerli et al.[29] | Case-Control (R) | Male and female high school athletes (diverse sports) | 23 cases with ACL rupture, 115 matched controls | ImPACT (visual memory, verbal memory, visual motor speed, RT), incidence of ACL rupture | Higher RT (+10.3%), lower processing speed (-10.3%) as well as lower verbal memory (-7.7%) scores in injured athletes. No difference in visual memory. |

Yrs = years, P = prospective, R = retrospective, NR = not reported, ACL = anterior cruciate ligament, ms = miliseconds, OR = Odds Ratio, RR = relative risk, RT = reaction time, *: no effect size could be extracted.

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**Tab.2.** Methodological quality of the included studies as rated with the modified checklist

| Study | Objective stated | Sample characteristics clearly described | Outcomes clearly described | Findings clearly presented | Variability estimates reported | Actual probability values given | Invited subjects representative | Participating subjects representative | Data dredging | Statistical tests appropriate | Identical time periods | Outcome measures accurate | Adjustment for confounders | Recruitment from the same population | Recruitment during same time period | Power | Sum score |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Taimela et al.[21*] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| Dvorak et al.[22] | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 10 |
| Swanik et al.[28] | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 12 |
| Wilkerson et al.[23] | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 12 |
| Faltus et al.[24] | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 10 |
| Wilkerson et al.[25] | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 15 |
| McDonald et al.[27] | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 13 |
| Ha et al.[26] | 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 11 |
| Eyerli et al.[29*] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |

1: criterion met (1 point awarded), /: criterion not met (0 points awarded), n/a: not applicable (not included in composite score), *: abstract/no full-text available for rating

## Figure legends

**Fig. 1.** PRISMA chart displaying the flow of the literature search

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**Confirmatin on Ethical compliance:** Does not apply

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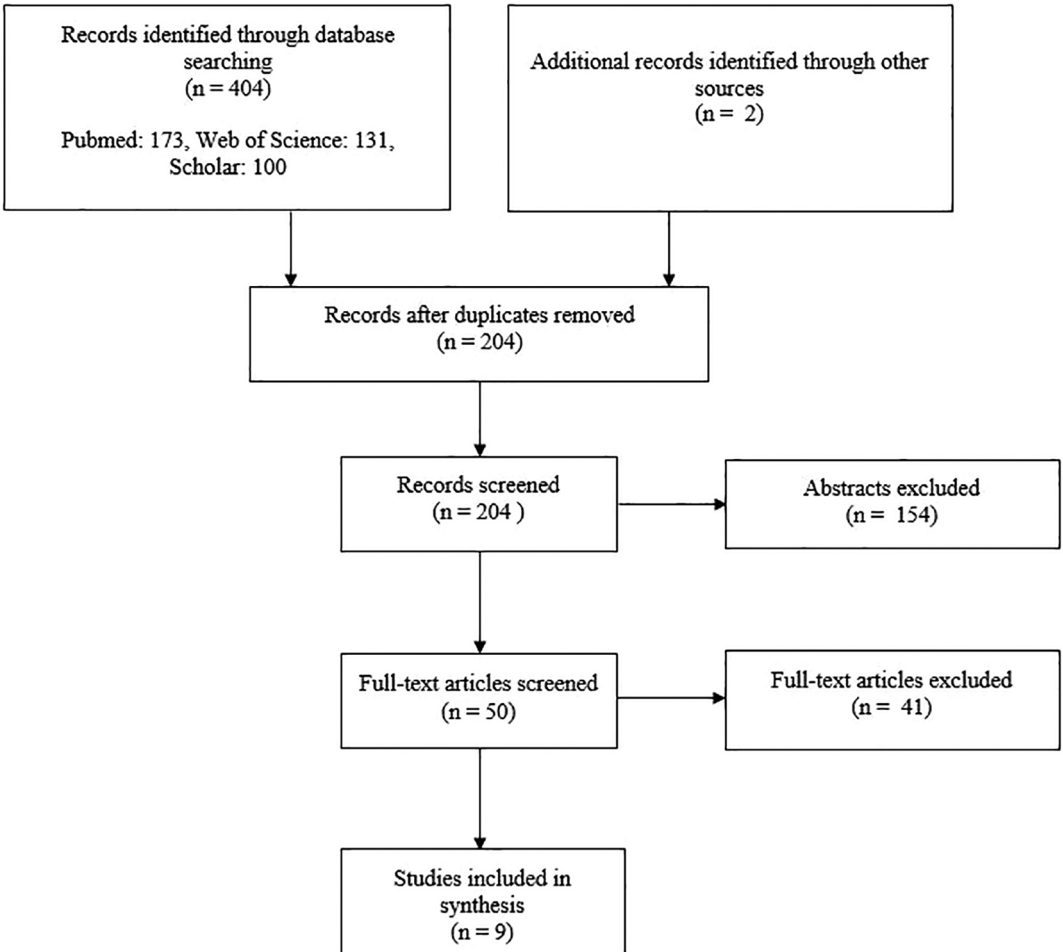
Records identified through database searching
(n = 404)
Pubmed: 173, Web of Science: 131, Scholar: 100
Additional records identified through other sources
(n = 2)
Records after duplicates removed
(n = 204)
Records screened
(n = 204 )
Abstracts excluded
(n = 154)
Full-text articles screened
(n = 50)
Full-text articles excluded
(n = 41)
Studies included in synthesis
(n = 9)

Figure 1